Agentic Coding

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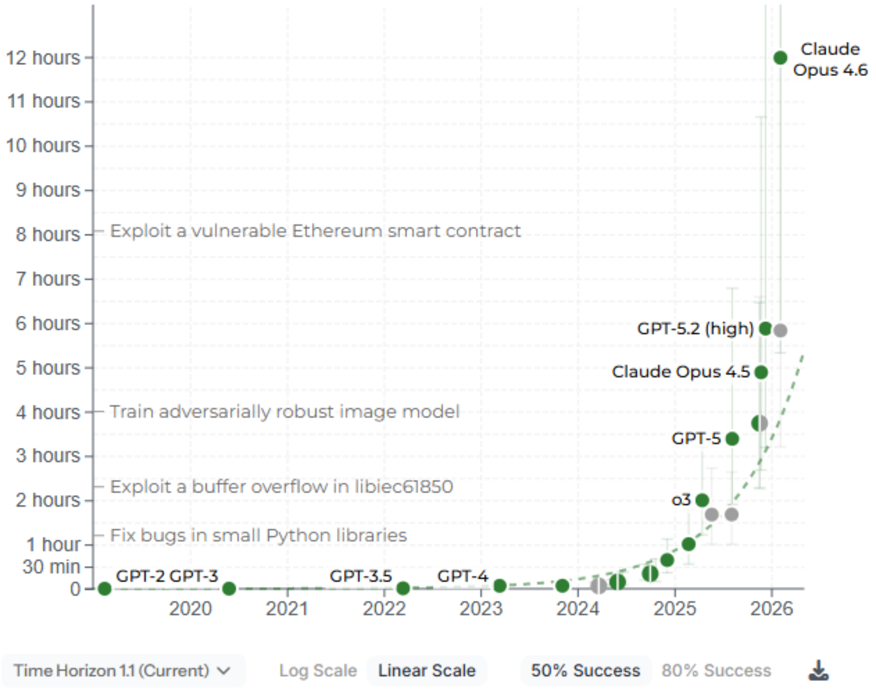
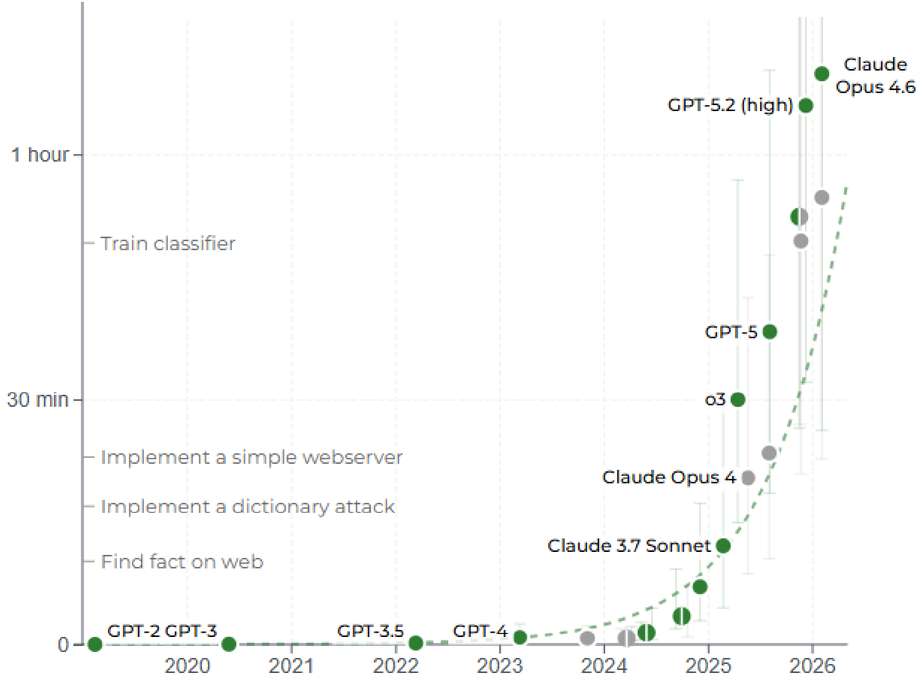
I don't think I've typed like a line of code probably since December, basically, which is an extremely large change.

Andrej Karpathy, AI Researcher

Agents

Coding Agents: Ability to perform long tasks





Source: https://metr.org/

About me

Bert Gollnick

Dipl.-Ing. Aerospace Engineering

M. Sc. Economics

Aerodynamics Engineer

Data Analyst

Senior Data Scientist

Gollnick Data Solutions - Founder

What you will learn

Three things you'll be able to do after this presentation

1. You understand what sets Coding Agents apart from autocomplete and chatbots—and why this is in a league of its own, especially right now.
2. You're familiar with the key tools on the market (CLI, IDE, browser) and know which one suits which use case.

Agenda

What we will cover

1. What is an Agent? — and how does it work
2. Coding Agents on the market
3. Agentic Coding Workflow
4. Claude Code Deep Dive
5. The 7 Levels of Agentic Coding
6. Socio-economic impact

What is an Agent?

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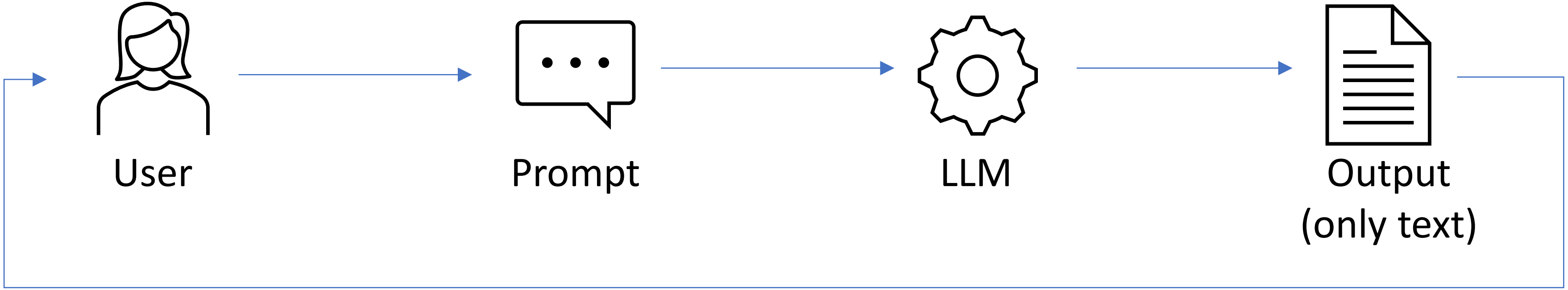
What is an Agent?

Chatbot



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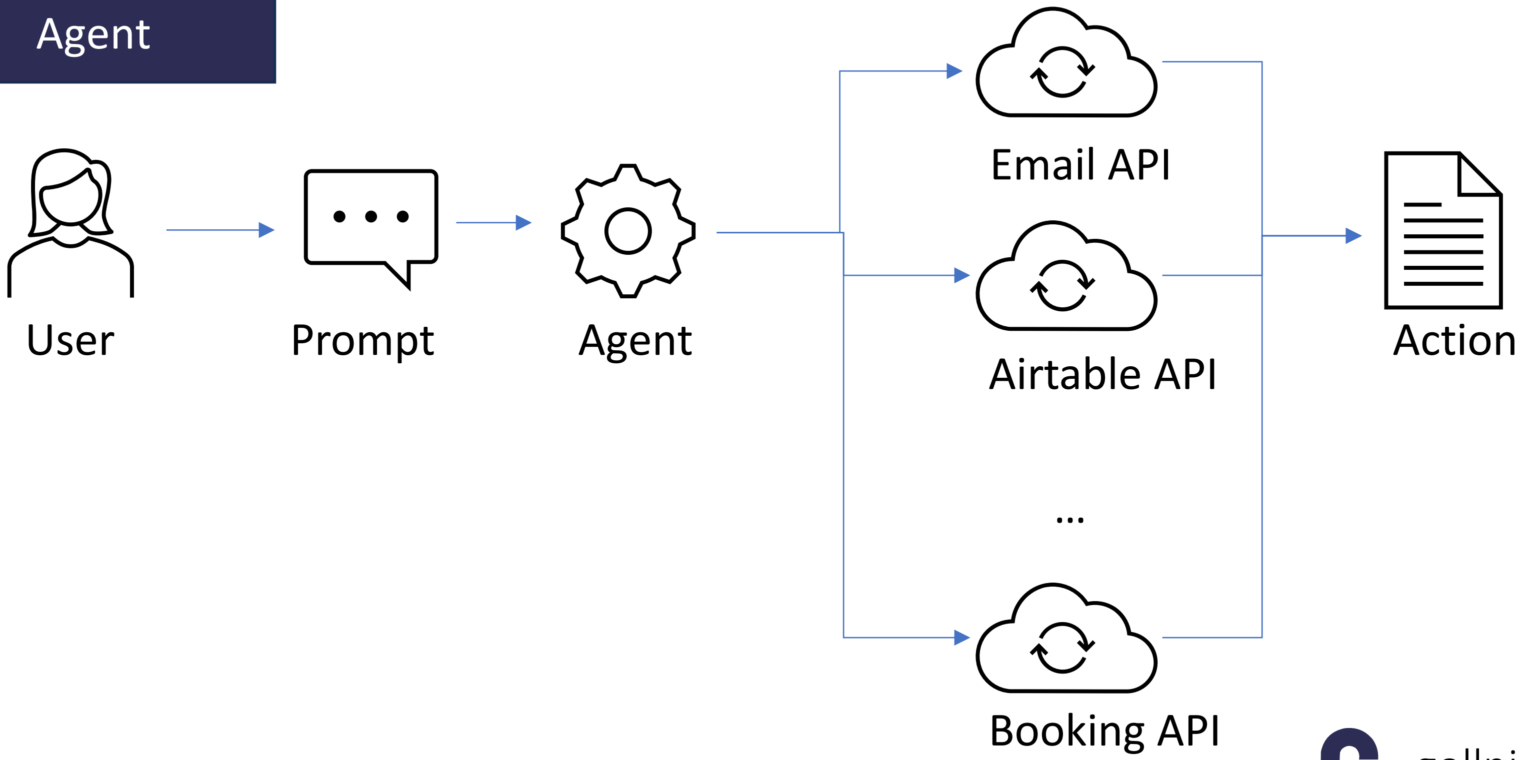
What is an Agent?

Agent



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What is an Agent?

- Perceives Environment
- Takes Actions autonomously
- To achieve a defined goal
- Improve through learning

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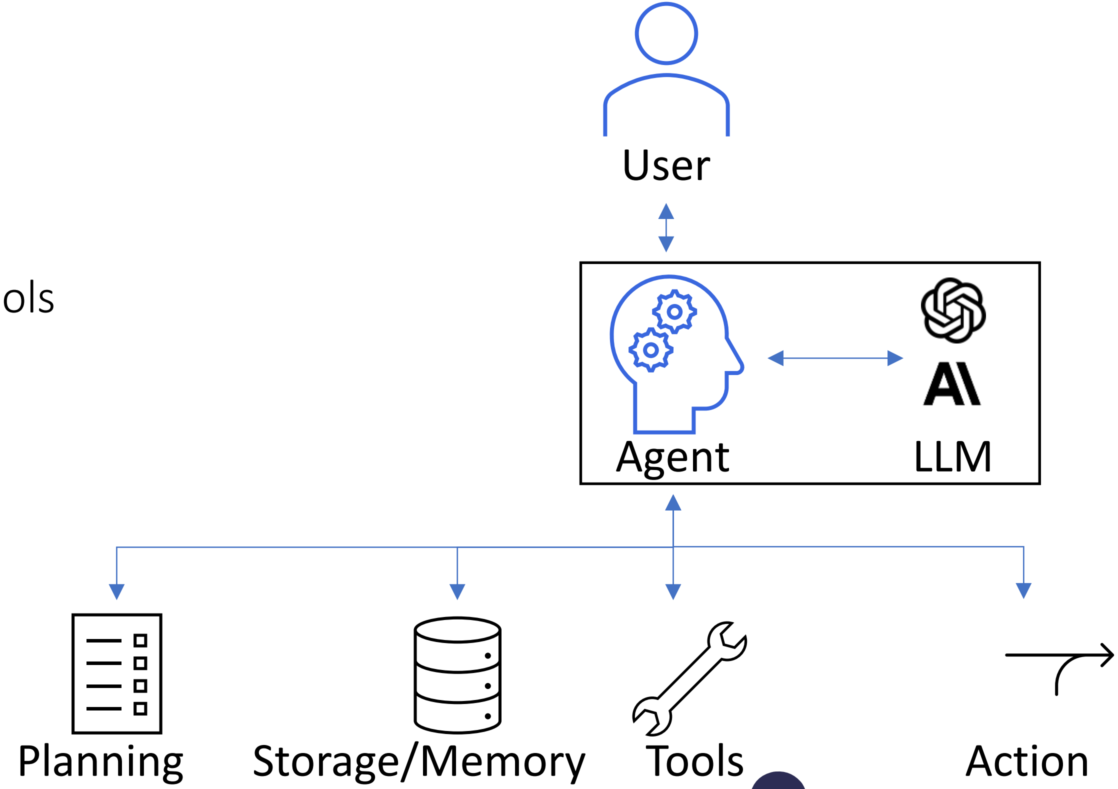
What is an Agent?

- LLM apps execute tasks
- core element: agent
- uses planning, memory, and tools
- can perform actions
- is an expert in its field



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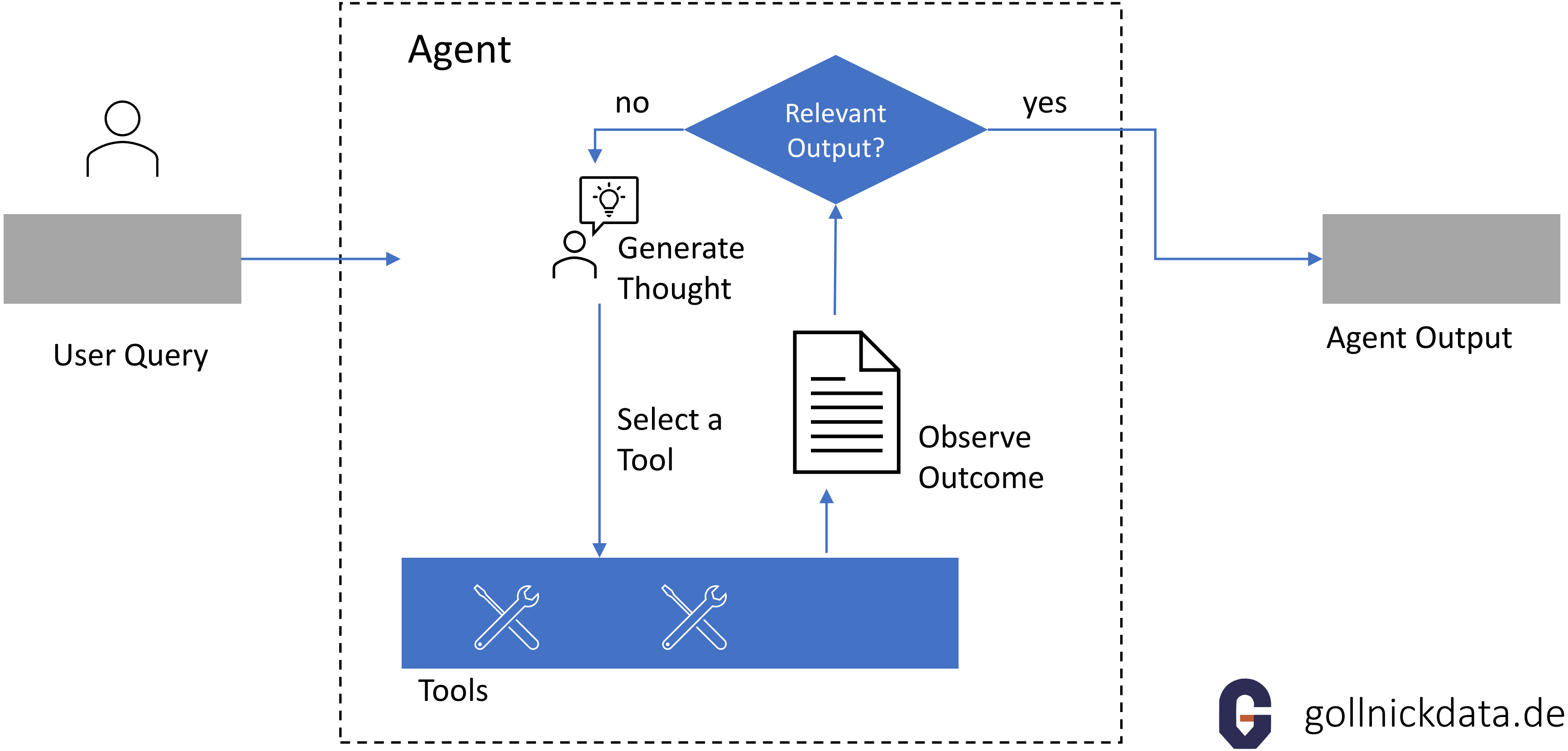
How does an Agent work?



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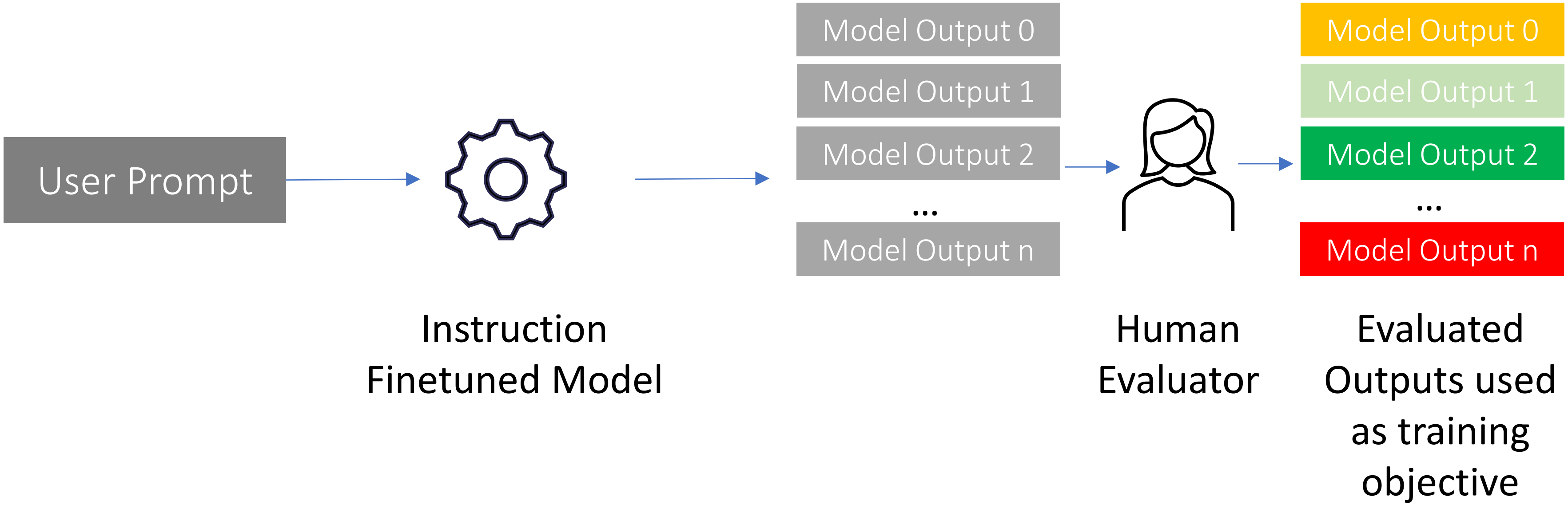
Large Language Models

Training Process: Reinforcement Learning from Human Feedback



The model predicts what a human would answer.

Large Language Models

Training Process: Reinforcement Learning from Verifiable Rewards (RLVR)

User Prompt

Instruction Finetuned Model

Model Output 0

Model Output 1

Model Output 2

...

Model Output n

Verifiable Algorithm, e.g. code compiler, mathematical formula

Model Output 0

Model Output 1

Model Output 2

...

Model Output n

Evaluated Outputs used as training objective

AI only gets rewarded for 100% correct answers

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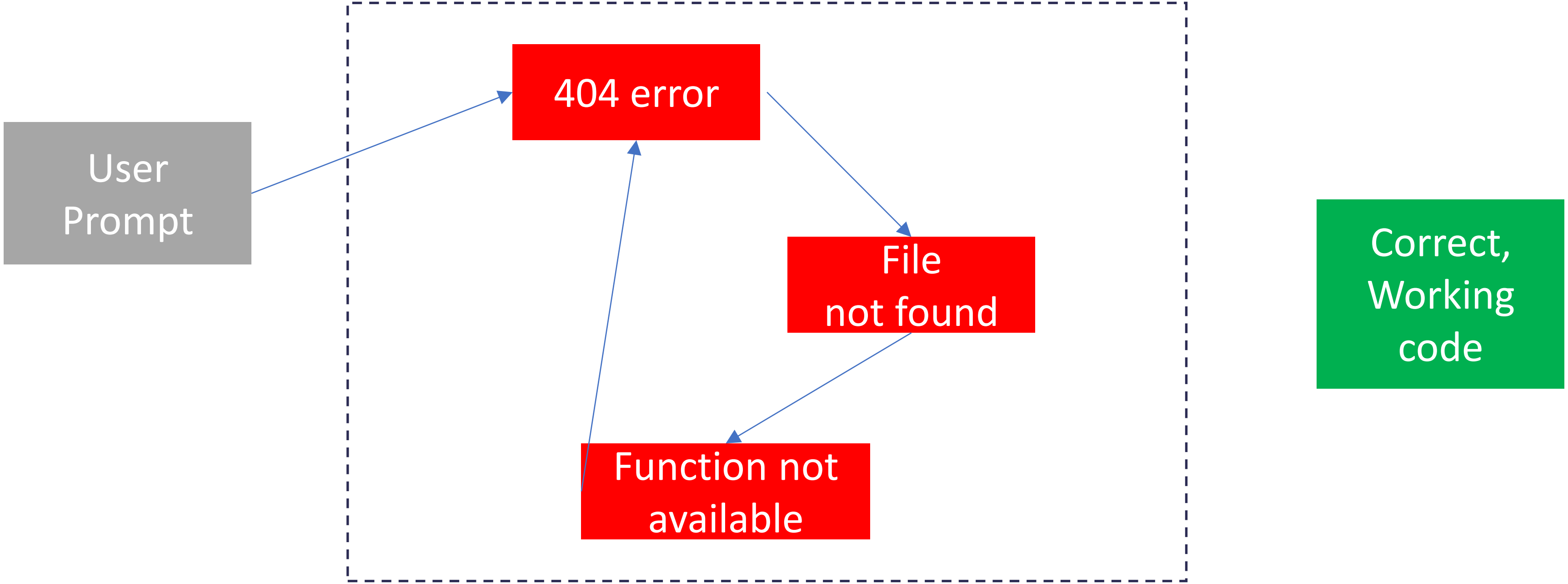
Issue: Semantic Drift



Errors are scattered and not clustered. Agent cannot differentiate poor from good solutions.

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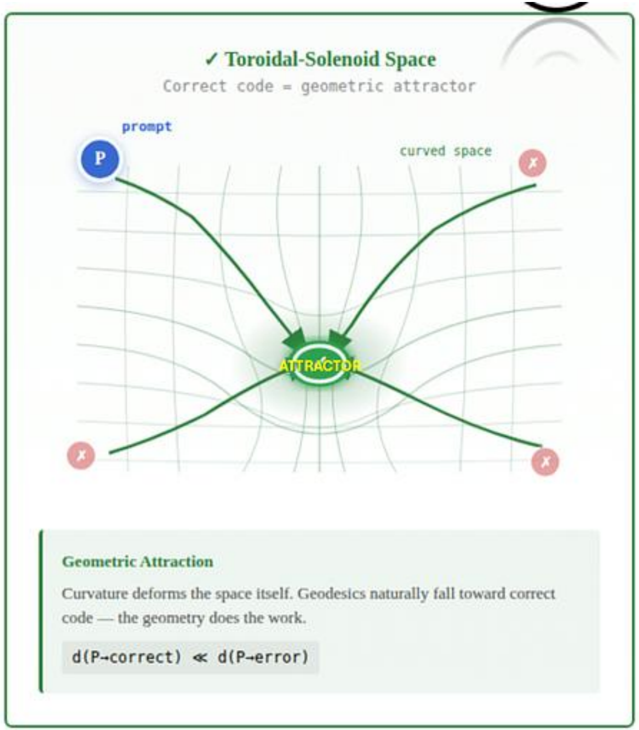
Solution: Curved Manifolds



When Models Manipulate Manifolds: The Geometry of a Counting Task

Wes Gurnee,* Emmanuel Ameisen,* Isaac Kauvar, Julius Tarng,

Adam Pearce, Chris Olah, Joshua Batson *†

Anthropic

*Equal contribution. †Corresponding author: joshb@anthropic.com

https://arxiv.org/html/2601.04480v1

- Errors are uphill
- Correct solutions are uphill

→ Coding agent is attracted by correct solutions

Sources: https://ai.gopubby.com/you-all-are-wrong-using-ai-for-programming-until-now-42a00e6a8e88

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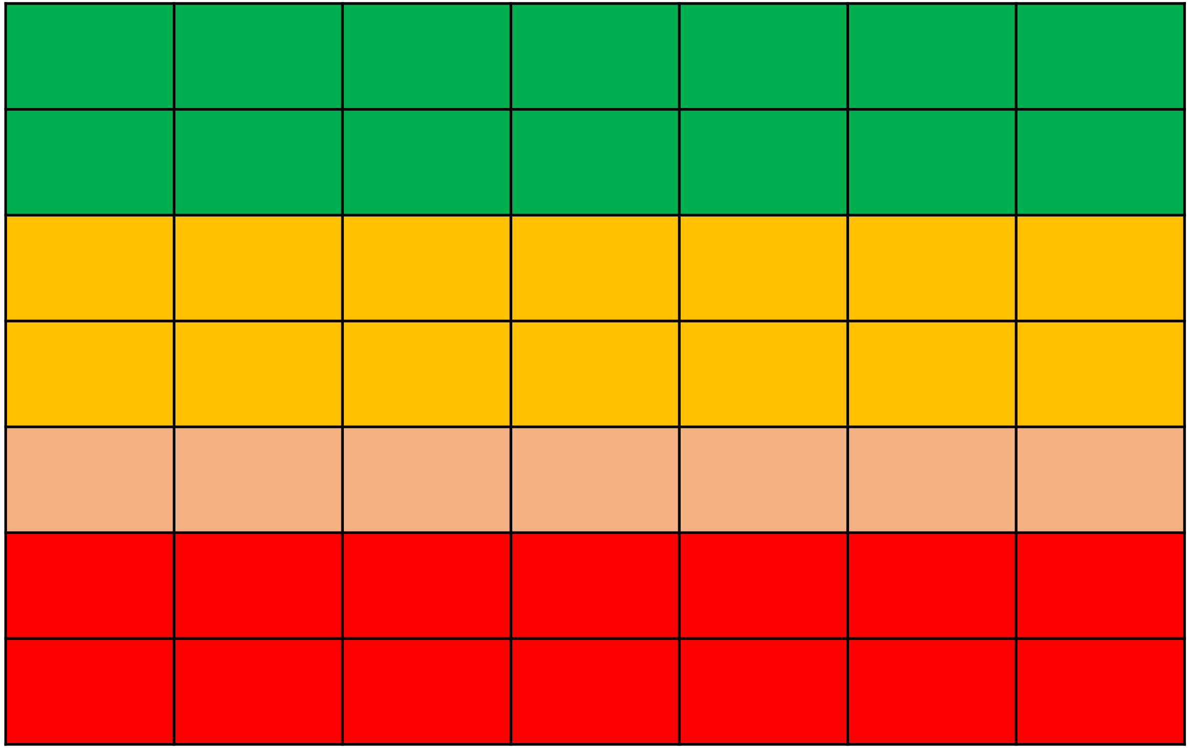
Context Rot: The needle in a haystack problem



Context Window

0-30%: peak quality

>30%: model skips details, makes shortcuts

>70%: hallucinations increase

>80%: earlier discussed requirements are forgotten

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Distractors: The Silent Failure Mode

Similar-but-wrong information pulled confidently into context poisons the entire run.

What Happens

- Retrieval surfaces near-matches that look authoritative
- Agent treats them as ground truth
- Downstream reasoning compounds the error
- Symptoms look like a model failure, not a context failure

Common Sources

- Outdated docs next to current docs
- Similar function names across modules (fetchUser vs getUser)
- Legacy APIs shadowing the new ones
- Deprecated patterns in Stack Overflow answers

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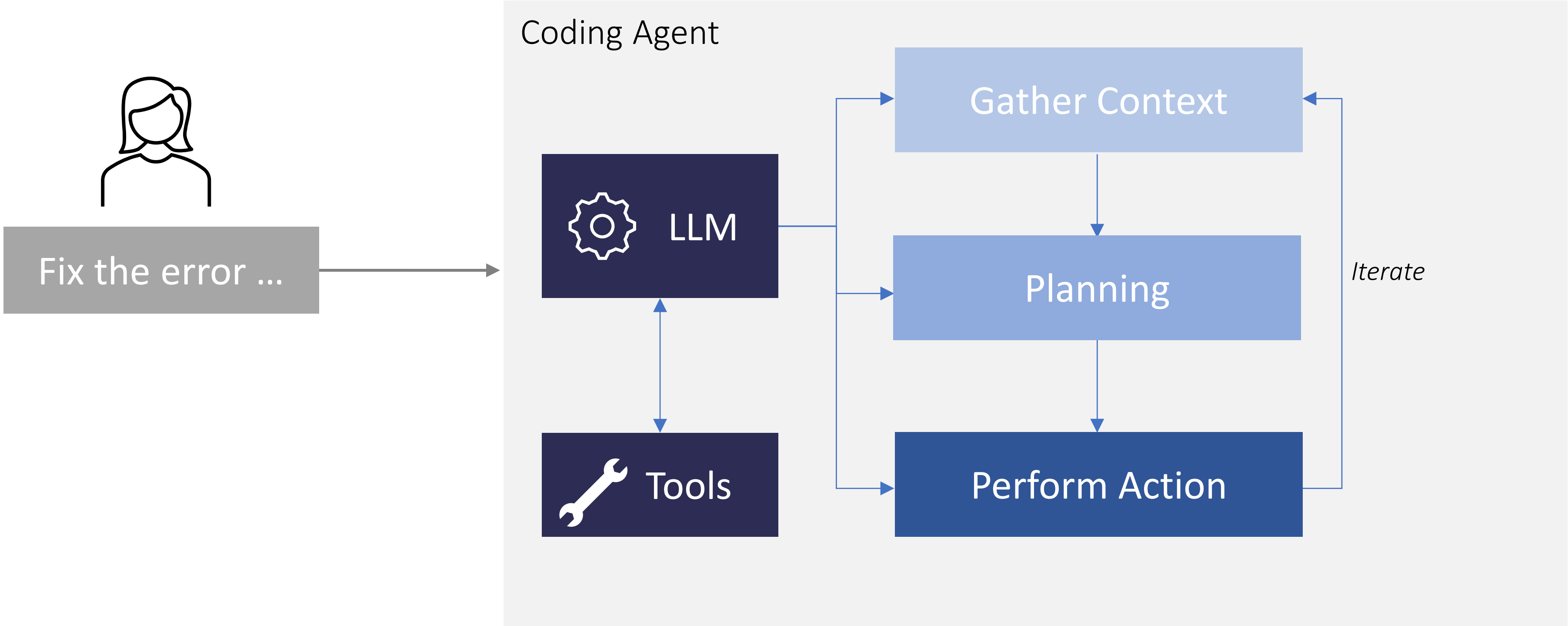
What is a Coding Agent?



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What is a Coding Agent?

Agent = LLM + Harness

Intelligence

- Reasoning
- Tool calling
- Token generation
- Pattern matching

everything else

- Context
- Memory
- Orchestration
- Feedback loops
- Tool wiring

Adapted from https://medium.com/ai-advances/harness-engineering-what-every-ai-engineer-needs-to-know-in-2026-0ab649e5686a

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Computer Analogy

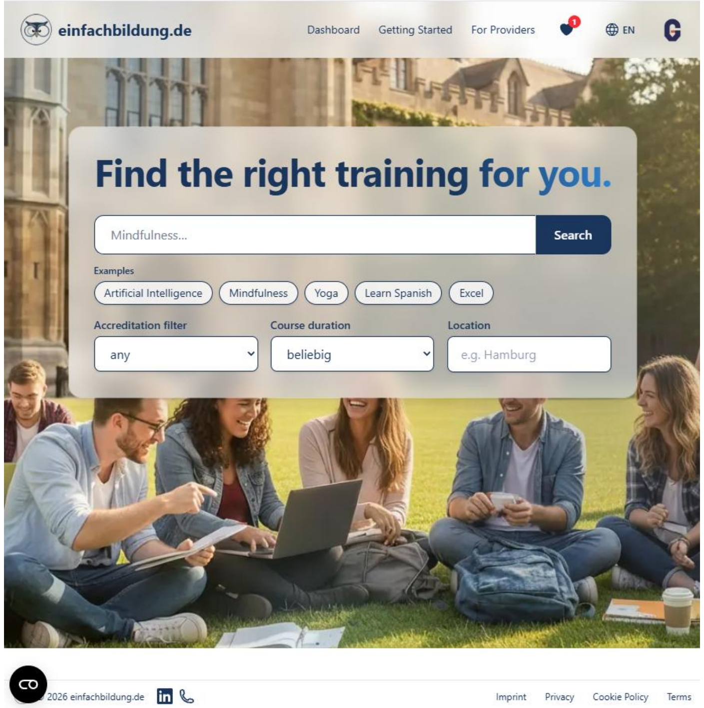
CPU	Model
RAM	Context Window
OS	Agent Harness
Application	Agent

adapted from Philipp Schmid, Cobus Greyling & AI,
URL: https://cobusgreyling.substack.com/p/the-rise-of-ai-harness-engineering

Claude Code

For what purpose do I use agentic coding

App-Development



Office Work

E	F	G	H	
Name der Gegenpartei	Cashflow-Kategorie	Cashflow-Unterkategorie	Rechnungsdokument	Kommentar
STRIPE TECHNOLOGY EUROPE, LIM	Verkaufserlöse	Online-Zahlungen	Rechnung_RE-2026-108.pdf	
Qonto	Bankgebühren	Sonstige Bankgebühren	20260429_qonto_gebühren.pdf	Qonto FX-Gebühr fü
VERCEL INC.	Technologiekosten	Hosting-Dienstleistungen	20260429_vercel.pdf	Vercel Pro April 202
STRIPE TECHNOLOGY EUROPE, LIM	Verkaufserlöse	Online-Zahlungen	Rechnung_RE-2026-106.pdf, Rechnung_R	Stripe-Auszahlung; S
MSFT * E0600Z87NE	Technologiekosten	Software-Lizenzen	20260427_Microsoft.pdf	Microsoft 365 (Rech
Bert Gollnick	Personalkosten	Gehälter		Gehalt April 2026 Be
Qonto	Bankgebühren	Sonstige Bankgebühren	20260425_qonto_gebühren2.pdf	Qonto FX-Gebühr; k
Qonto	Bankgebühren	Sonstige Bankgebühren	20260424_qonto_gebühren.pdf	Qonto FX-Gebühr; k
Qonto	Bankgebühren	Sonstige Bankgebühren	20260425_qonto_gebühren	Qonto FX-Gebühr; k

Tax Statements

Herzlich willkommen im Kurs 'Agentic Coding'

I

info@gollnickdata.de

An [redacted]

Anschreiben_Linus Ludwig.pdf
267 KB

Rechnung_RE-2026-112.pdf
239 KB

Sehr geehrter Herr [redacted]

ich freue mich Sie im Kurs begrüßen zu dürfen.

Invoices and emails

Coding Agents on the market

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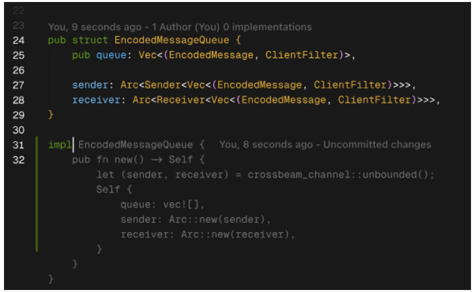
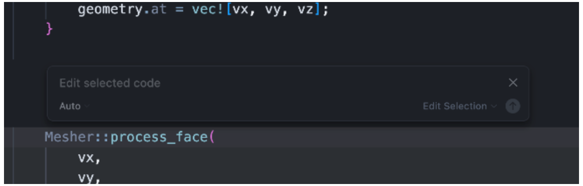
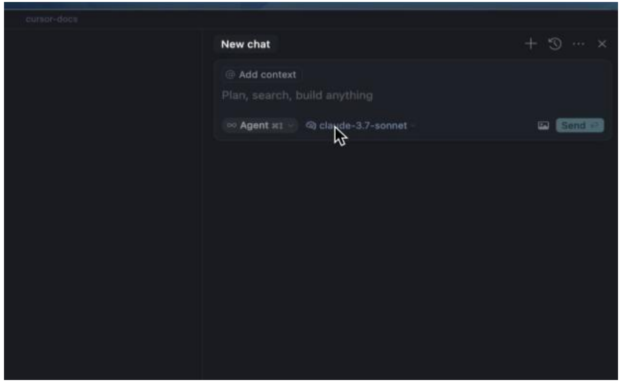
Coding Agents: Evolution

Auto Completion → Inline Complete → Complete code base







- Proposes automatically next code tokens

- Edits multiple lines
- Available in VS Code, Cursor, Windsurf

- Accesses complete code base
- Performs edits

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Vibe Coding vs. Agentic Coding

Applications of Vibe Coding

Rapid Prototyping

UI/UX Development

Learning & Exploration

Debugging

Applications of Agentic Coding

Enterprise Software

Infrastructure as Code

Data Pipelines

Maintenance

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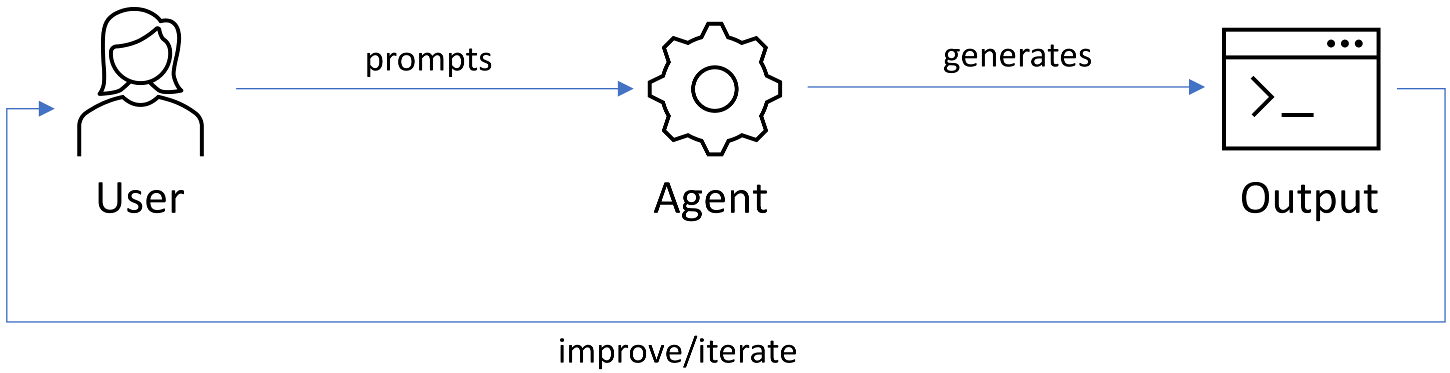
Vibe Coding vs. Agentic Coding

Vibe Coding



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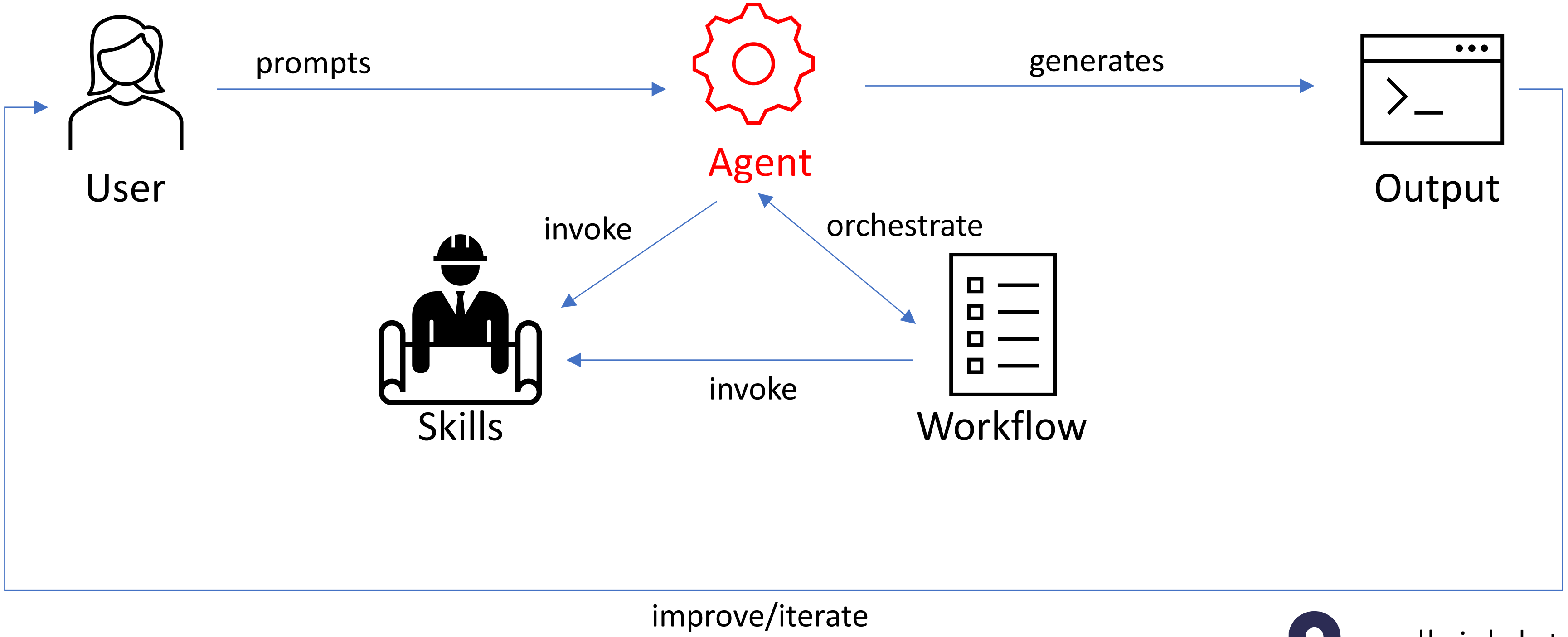
Vibe Coding vs. Agentic Coding

Agentic Coding



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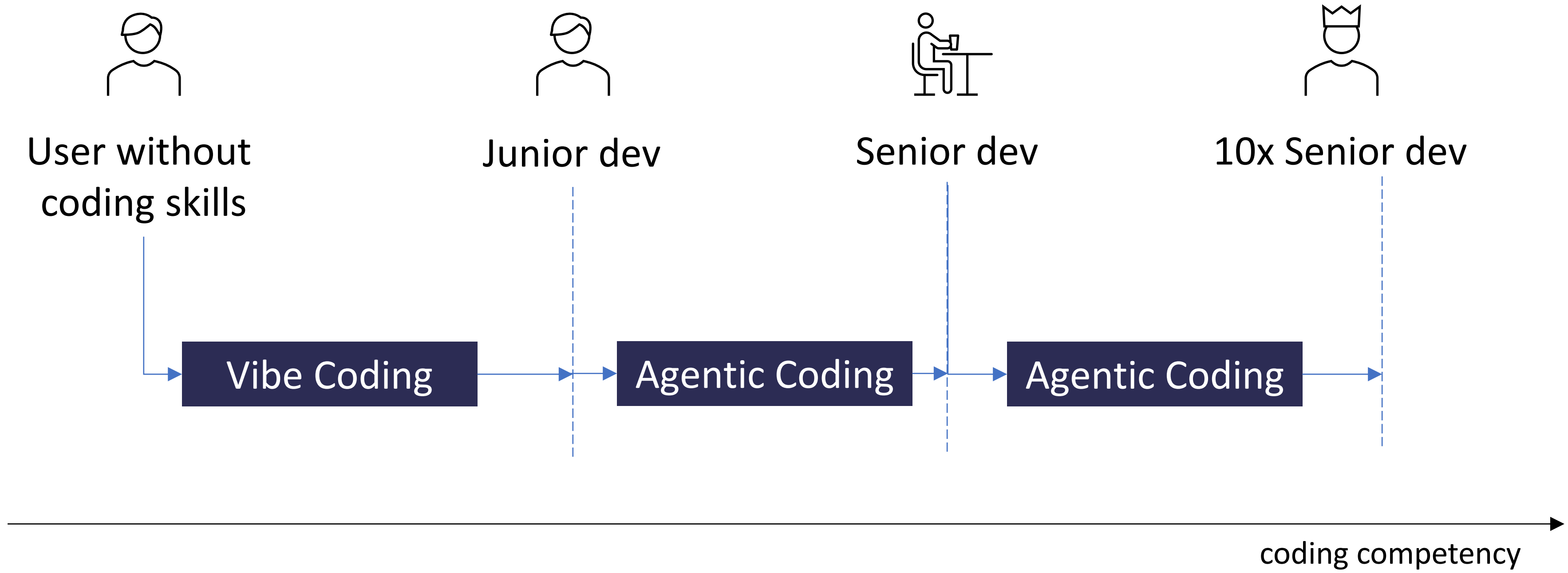
Vibe Coding vs. Agentic Coding



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(Vibe) Coding Agents: Browser-Based

- Browser-based
- freemium
- https://replit.com/

- Browser-based
- freemium
- https://lovable.dev/

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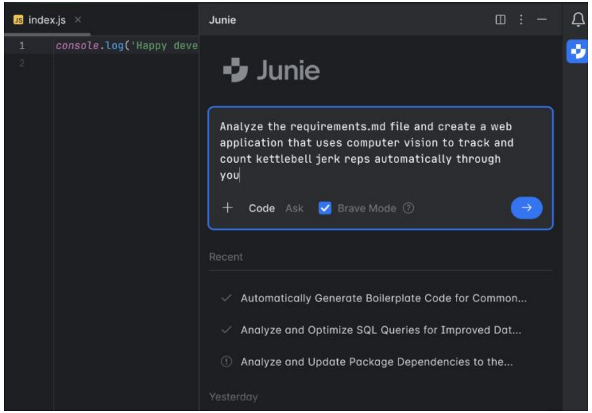
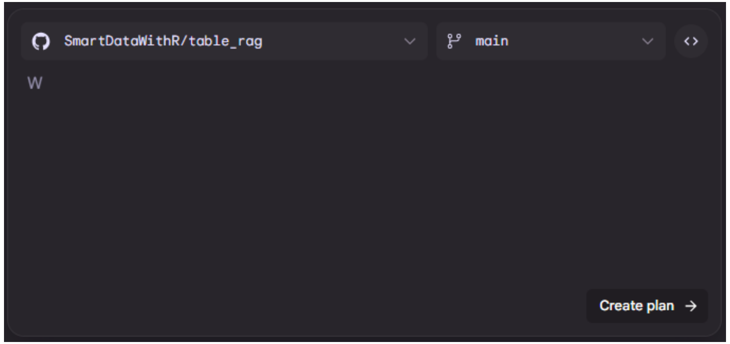
Coding Agents: Browser-Based

Junie



- Developed by Jetbrain
- Available in Jetbrain-IDEs (e.g. PyCharm)

Jules



- Browser-based
- Developed by Google
- Integrates with GitHub

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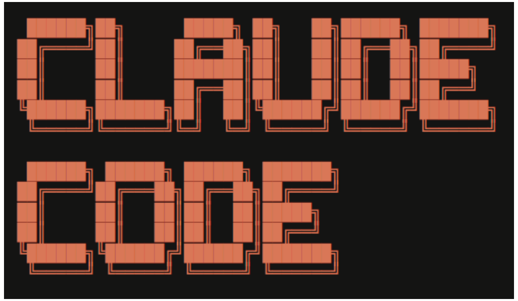
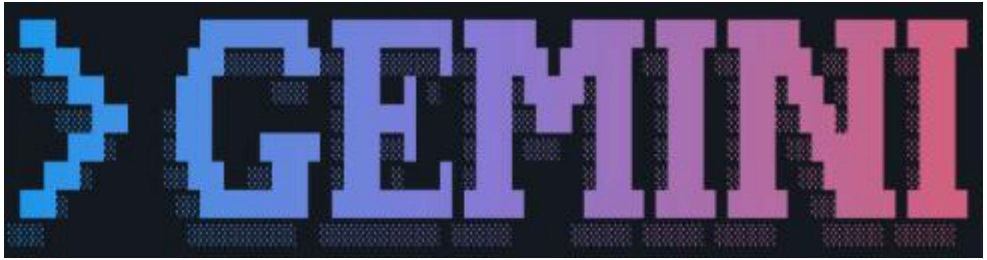
Coding Agents: Terminal



- Works from command line
- Can work with your complete code base
- Install: npm install -g @anthropic-ai/claude-code
- Available for claude subscribers or API-use



- Source
- Can work with your complete code base

- Install: npm install -g @google/gemini-cli
- Available via API key or Google SignIn
- 100 Gemini 2.5 Pro requests free per day!

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Coding Agents: Terminal

- Source
- Open Source
- Can work with your complete code base
- Less secure, but lets you look into Claude Code

Install: git clone https://github.com/ultraworkers/claw-code
cd claw-code/rust
cargo build --workspace

- Available for any LLM Provider

- Source
- Can work with your complete code base

- Install: curl -fsSL https://opencode.ai/install | bash
- Available via API key

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Coding Agents: Terminal

Codex

- Developed by OpenAI
- Included in ChatGPT Plus, Pro, Business, Edu, Enterprise, or via API-key
- Works from command line

Mistral Vibe

Keep shipping.

- Developed by Mistral
- Released under Apache 2 license

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Coding Agents: IDE

<u>Cursor</u>

- 20$/month
- Suitable for
  - Experienced devs, large codebase
- Makes Claude 4 available

<u>Windsurf</u>

- 15$/month
- Suitable for
  - Beginner, vibe-coding, personal projects
- Develop web apps quick and without configuration effort

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Coding Agents: IDE

CLAUDE CODE

CLAW Code

CLI

opencode

GEMINI

IDE

CURSOR

replit

Lovable

Junie

jules

User Experience Level

Beginner

Intermediate

Senior-Developer

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Coding Agents: Arena

Source: https://app.daily.dev/agents/arena

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Coding Agents: Arena

	TOOL	D-INDEX	24H VOL	SENTIMENT	MOMENTUM	7D TREND
	Claude Code	235	786	62	+5%	
	Codex	182	610	62	+9%	
	Cursor	141	608	56	-18%	
4	Antigravity	76	341	53	+10%	
5	OpenCode	74	256	58	-34%	
6	Copilot	33	125	54	+6%	
7	Windsurf	23	62	56	+14%	

	TOOL	D-INDEX	24H VOL	SENTIMENT	MOMENTUM	7D TREND
	Gemini	145	699	53	+10%	
	Claude Opus	133	432	61	+20%	
	Qwen	121	325	63	-16%	
4	GPT-5	111	543	53	-2%	
5	DeepSeek	93	446	53	+4%	
6	GPT Codex	89	267	62	-10%	

Source: https://app.daily.dev/agents/arena

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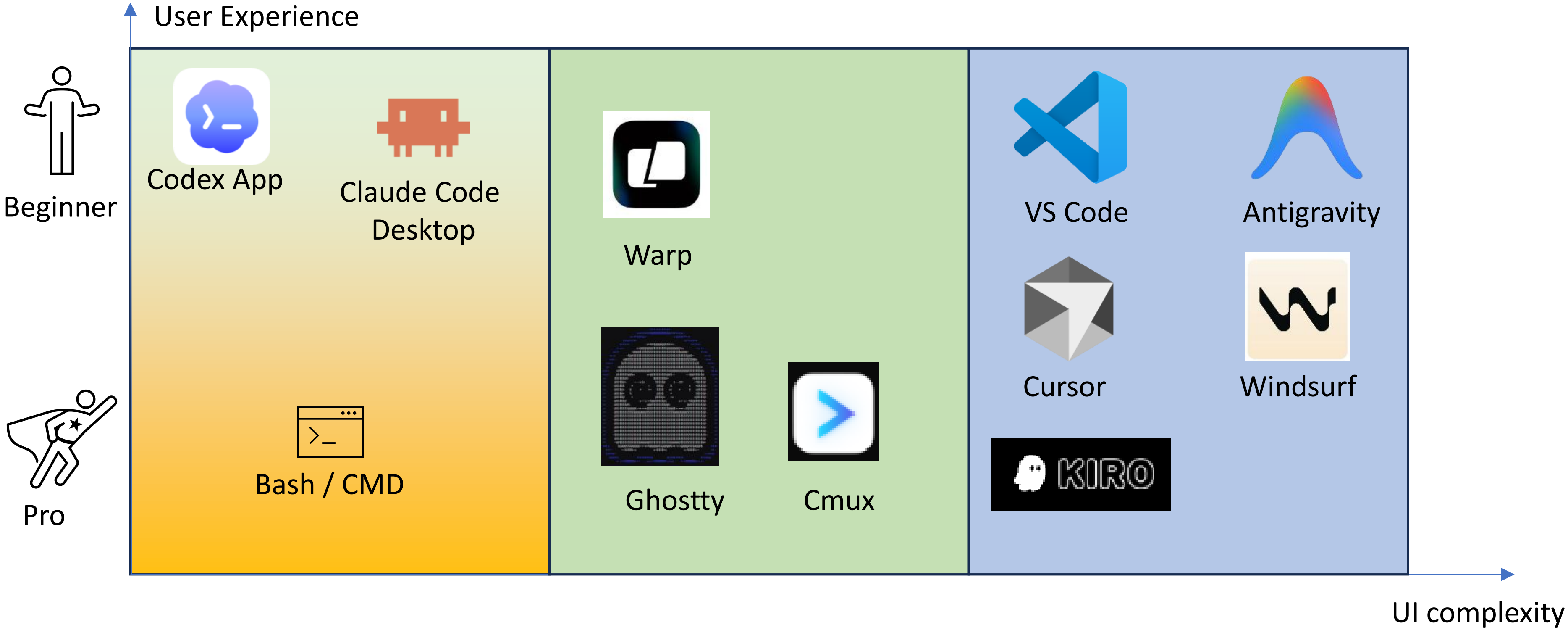
Coding Agents: Terminal vs. Standalone vs. IDE



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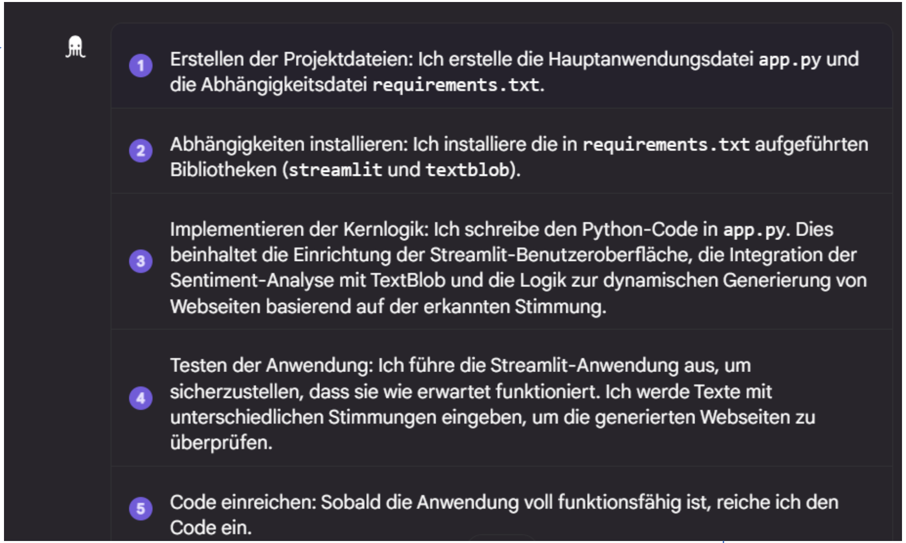
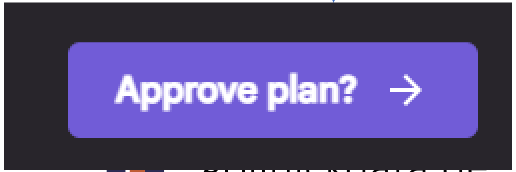
Coding Agents: Exercise

User Prompt

Diese Anwendung liest einen kurzen Text ein und bewertet dessen emotionale Stimmung (z. B. positiv, negativ, neutral) mithilfe von Sentiment-Analyse. Basierend auf dem ermittelten Vibe generiert die App eine Webseite. Beispielsweise könnte ein positiver Text eine helle, freundliche Webseite mit vielen leuchtenden Farben und optimistischen Sprüchen erzeugen, während ein negativer Text eine düstere, minimalistische Seite mit gedämpften Farben und melancholischen Zitaten generiert. Der Code wird in Python geschrieben und streamlit für die App verwendet.

jules





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Coding Agents: Exercise

jules

app.py requirements.txt

/app.py

```python
import streamlit as st
from textblob import TextBlob

def generate_website(sentiment):
    """Generiert eine Webseite basierend auf der Stimmung."""
    if sentiment == "positiv":
        st.markdown("""
        <style>
        .container {
            background-color: #ffecb3;
            color: #333;
            padding: 20px;
            border-radius: 10px;
            text-align: center;
            font-family: 'Arial', sans-serif;
        }
        h1 {
            color: #ff8f00;
        }
        </style>
        <div class="container">
            <h1>Ein sonniger Tag! 🌞</h1>
            <p>Die Stimmung ist wunderbar positiv. Hier ist ein Lächeln für dich! 😊</p>
            <p><i>"Behalte dein Gesicht immer im Sonnenschein - und Schatten werden hinter dic
        </div>
        """, unsafe_allow_html=True)
    elif sentiment == "negativ":
        st.markdown("""
        <style>
        .container {
            background-color: #424242;
            color: #f5f5f5;
            padding: 20px;
            border-radius: 10px;
            text-align: center;
            font-family: 'Georgia', serif;
        }
        h1 {
            color: #9e9e9e;
```

app.py requirements.txt

/requirements.txt

```
streamlit
textblob
```

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Coding Agents: World before Agents

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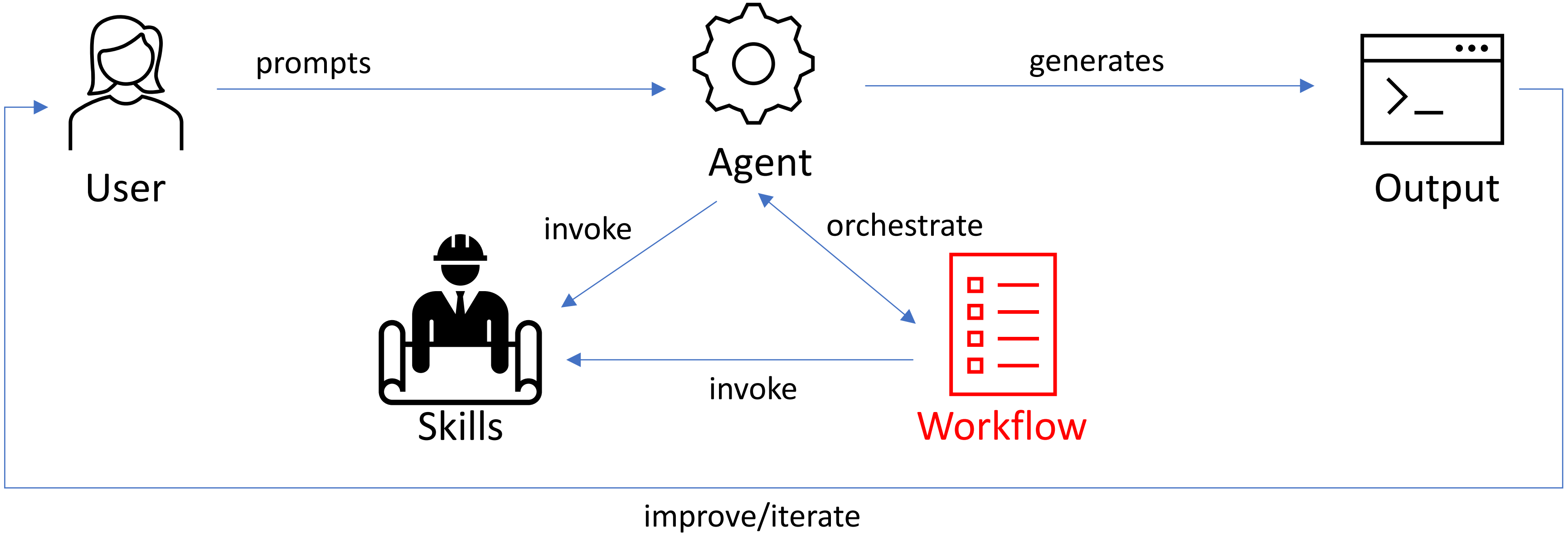
Agentic Coding: Workflow

Agentic Coding



The difference between vibe and agentic coding is discipline.

Agents

Karpathy's Four Principles

How to steer an agent: be concrete, be incremental, be skeptical.

The Principles

- Think before coding — write the plan in prose first
- Simplicity first — smallest change that moves the goal
- Surgical changes — no touching unrelated code
- Goal-driven — validate against the target, not the vibe

Why They Matter

- Agents fail silently when goals are vague
- Scope creep compounds across turns
- Broad edits explode review & rollback cost
- Without validation, success is indistinguishable fr om hallucination

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Validation Loop as First-Class Concern

Meta Staff Engineer insight: build/test/lint isn’t an afterthought — it’s the product.

The Loop

- 1. Agent proposes a change
- 2. Validation runs (build, test, lint, typecheck)
- 3. Failures feed back as concrete errors
- 4. Agent iterates until green — or asks for help

Design Implications

- Fast feedback beats smart prompts — optimize f or loop speed
- Red-green signals must be unambiguous
- If validation is slow or flaky, the agent gets lost
- Invest in your test suite before investing in your pr ompts

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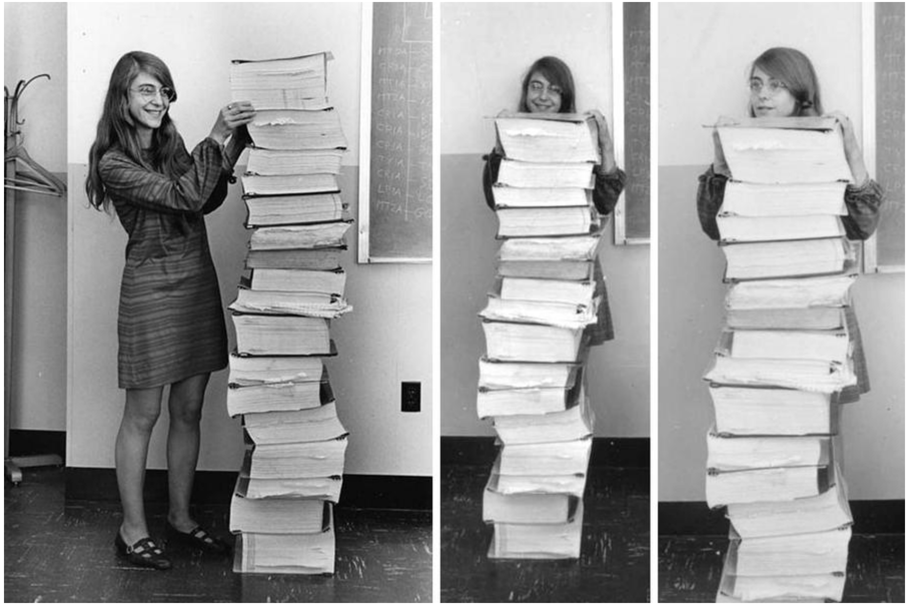
Agentic Coding: Workflow

Why?



- Apollo program did have any error margin
- More critical environments require stricter guidelines and processes
- NASA developed rules to follow
- Software devs use gateways and pipelines

Apollo guidance software

Source: https://news.mit.edu/2016/scene-at-mit-margaret-hamilton-apollo-code-0817

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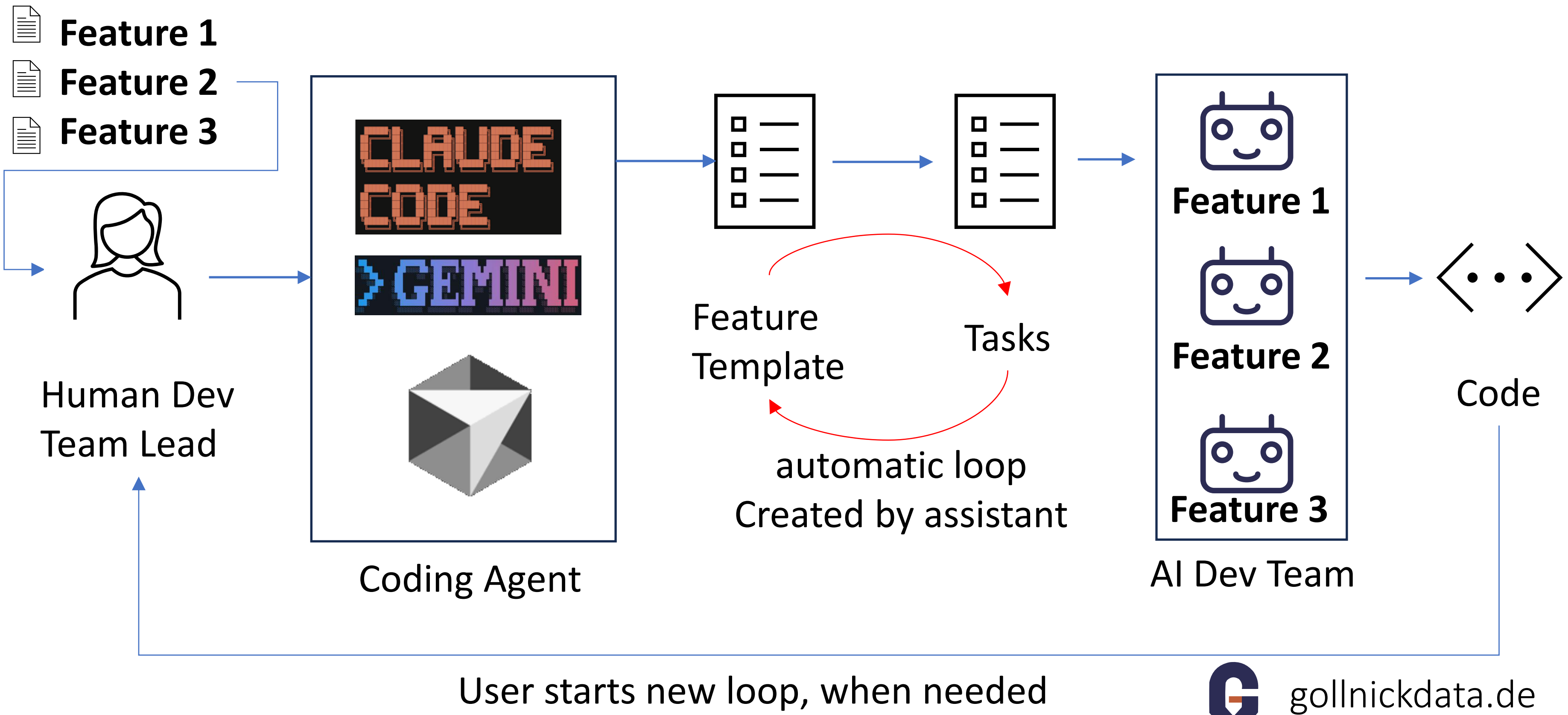
Agentic Coding: Workflow



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Agentic Coding: Workflow

BMAD

- Breakthrough Method for Agile AI-Driven Development
- 12+ Personas
- Simulates agile development team
- 4-phase agile cycle with Analysis, Planning, Solution, Implementation
- 30+ workflows

SpecKit

- CLI commands
- „specifications don‘t serve code, code serves specification
- Gated process: constitution, spec, plan, tasks, implement

GSD

- Get Shit Done
- Specialised subagents: researcher, planner, executor, verifier
- Creates artifacts like spec.md, plan.md, research.md, …

Superpowers

- Forces structured dialogue between developer and AI agent
- Gated pipeline: brainstorm, TDD, implement, review
- Quality enforcement

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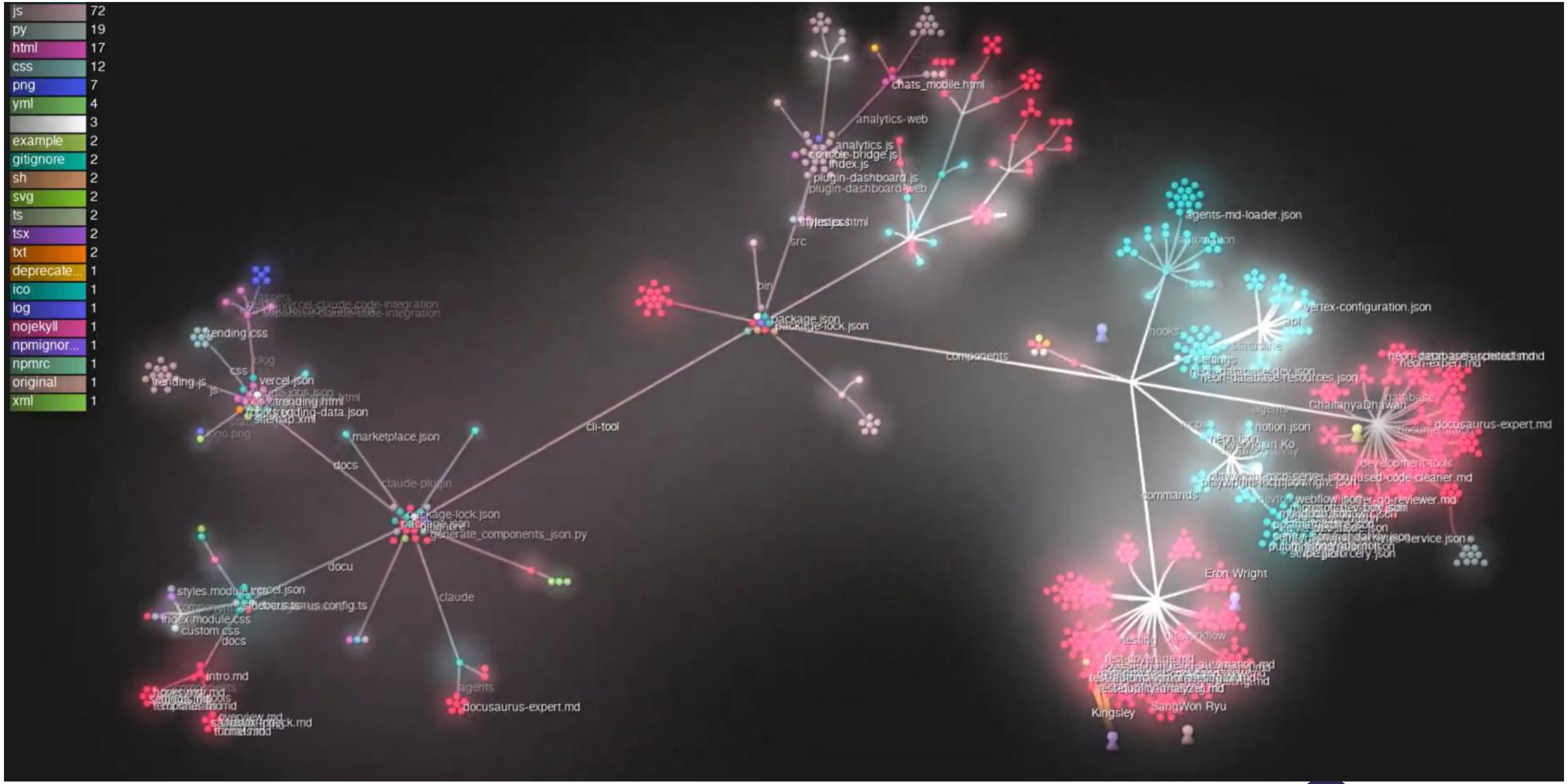
Coding Agents: Claude Code Visualisation



Source: https://x.com/dani_avila7/status/1978510902740877344

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Capabilities & Design Patterns

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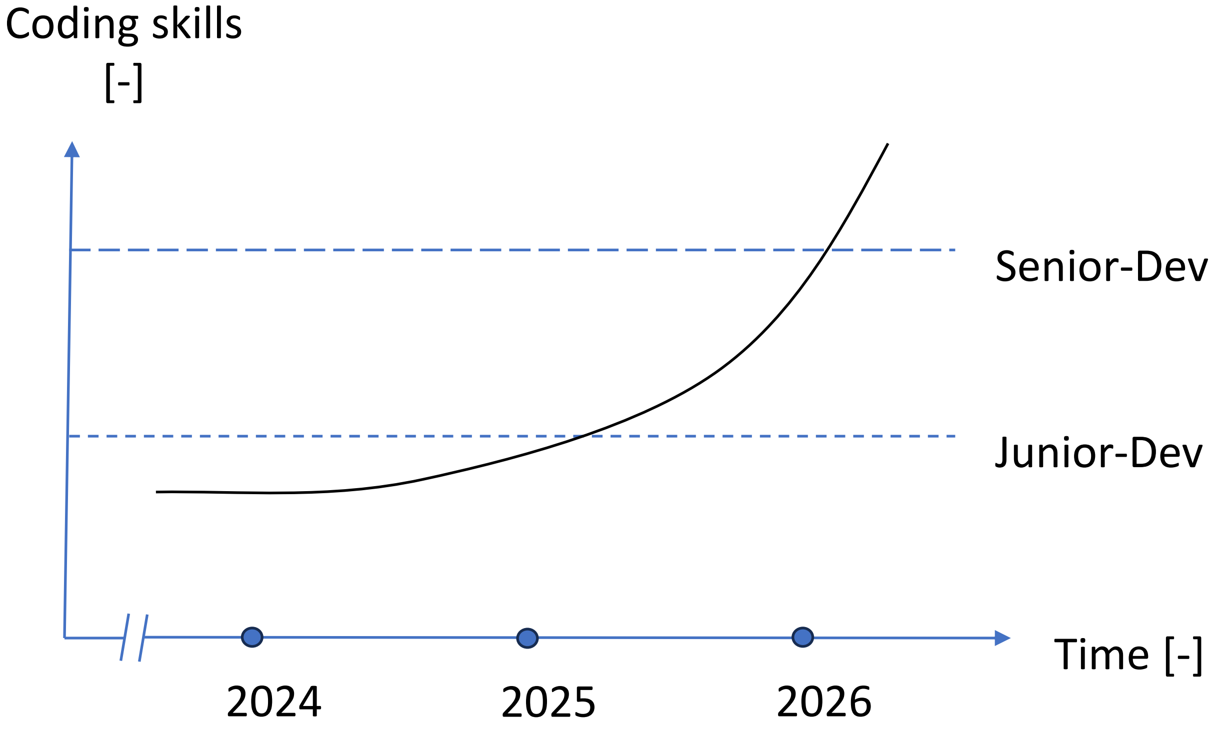
Coding Agents: Capabilities



Source: own graphic

Agents

Agentic Design Patterns (Anthropic)

Prompt Chaining

Sequentielle LLM-Aufrufe.
Output N → Input N+1.

Routing

Classifier erkennt Task-Typ.
Leitet an spezialisierten Agent weiter.

Parallelization

Mehrere LLM-Aufrufe gleichzeitig.
Ergebnisse werden zusammengeführt.

Orchestrator-Worker

Zentraler Agent zerlegt komplexe Tasks und delegiert an spezialisierte Worker.

Evaluator-Optimizer

Generator erstellt Output.
Evaluator iteriert bis zur Zielqualität.

Source: Anthropic, Building effective agents (2024)

Claude Code:
Introduction

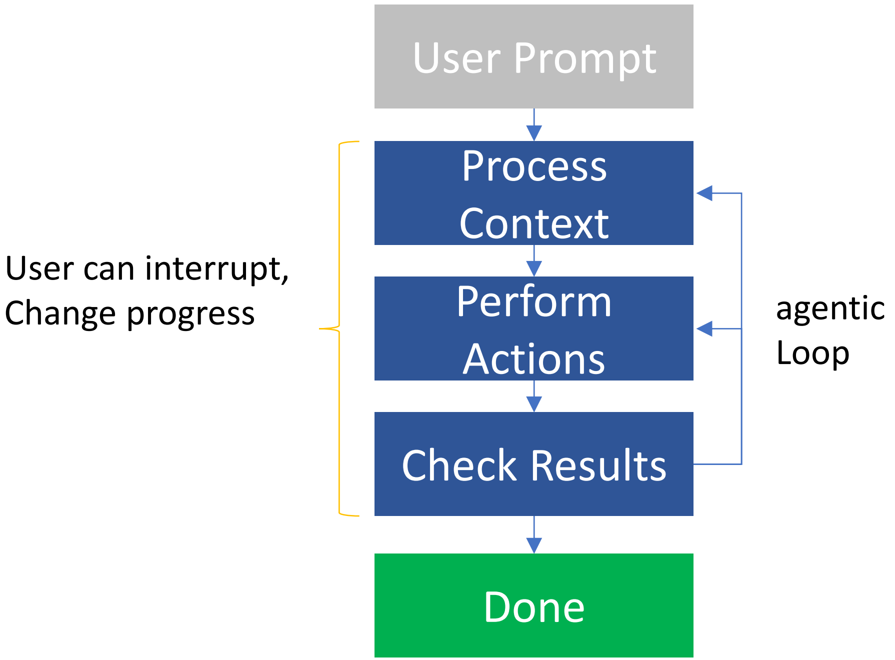
Claude Code

What is it?

An AI-Coding-Tool, that runs in the terminal.

Core Functions

- Develop features based on descriptions
- Debug
- Navigate in Codebase
- Manage Git-Workflows



Claude Code can edit files directly and perform commands.

Claude Code

Installation and Setup

Requirement

- Claude Subscription (Pro/Max/Teams/Enterprise)
- Or: API Access via Claude Console

Installation

Available for MacOS, Linux, Windows

https://code.claude.com/docs/en/setup

Add to path variable

Start

Run claude in Terminal

On first start: login required

Claude Code

Installation and Setup

Demo

Installation

- https://code.claude.com/docs/en/quickstart

Nutzung via API KEY

- Only relevant for participants using my API key
- Set Environment variable ANTHROPIC_API_KEY on system level
- Powershell: function claude-low { & claude --model claude-3-5-haiku-latest $args }
- Sets the model to cheap option

Claude Code

Change CLAUDE CODE to run with custom LLM

Since Ollama v0.14 — no proxy, no translation layer

Claude Code talks directly to localhost:11434

1 · Point Claude Code at Ollama

Set three environment variables:

```
export ANTHROPIC_BASE_URL=
http://localhost:11434
export ANTHROPIC_AUTH_TOKEN=ollama
export ANTHROPIC_API_KEY=""
```

Then: claude --model qwen2.5-coder:14b

2 · Fix the KV-cache gotcha

Attribution header invalidates cache
→ ~90 % slower without this fix

Edit ~/.claude.json:

```
{
"CLAUDE_CODE_ATTRIBUTION_HEADER": 0
}
```

Recommended models

qwen2.5-coder:7b — 16 GB RAM
qwen2.5-coder:14b — sweet spot, 24 GB
qwen3-coder:30b — agentic, 48 GB RAM
devstral-small:24b — RTX 4090

Minimum: 32K context · 64K+ recommended

When this setup shines

DSGVO / KRITIS — code stays local
Air-gapped dev on customer premises
High-volume tasks without API costs
Experimentation with open models

Switch back: unset ANTHROPIC_BASE_URL

Claude Code

Permission Modes

Demo

plan	default	acceptEdits	auto
Read-only	Fragt jedes Mal	Edits ohne Frage	Classifier-überwacht
Claude erkundet, liest, plant — aber schreibt nichts	Schreib- und Bash-Operationen brauchen Bestätigung	File-Schreiben + ein paar Bash-Befehle (mkdir, mv, cp …)	Separates Modell prüft Aktionen vor Ausführung
Für Architektur-Diskussionen	Für sensible Codebases	Für aktives Coding	Team/Enterprise Research Preview

⚠ bypassPermissions

--dangerously-skip-permissions · keinerlei Prüfung · nur in Sandbox/Container/CI

← mehr Sicherheit — mehr Geschwindigkeit →

In jedem Modus gilt: Writes auf geschützte Pfade werden nie automatisch genehmigt

Claude Code

Workflow-Process

Start in Planning mode
(SHIFT + TAB + TAB)

> Think through how to add OAuth authentication to our app. Show the architecture, files we'll need, and implementation steps. Feel free to ask me questions.

Test driven development

> Write comprehensive tests for the OAuth flow - signup, login, token refresh, and error cases. Make sure they fail for now. Then, implement the OAuth system to make these tests pass.

Start in Edit or Auto-Mode
(SHIFT + TAB)

Claude Code

Installation and Setup

Demo

Folder structure

Empfohlene Projektstruktur

my-project/

- CLAUDE.md — Projekt-Kontext & System-Prompt
- .mcp.json — MCP-Server-Konfiguration (team-weit)
- .claude/
  - commands/ — Custom Slash-Commands
  - skills/ — SKILL.md-Dateien für Workflows — optional
    - code-review/ — eigener Review-Skill
    - deploy-checklist/ — Deployment-Workflow
  - agents/ — Subagent-Definitionen — optional
  - hooks/ — Lifecycle-Scripts — optional
- ai_docs/ — Referenz-Dokumente für Claude
  - PRD.md — Product Requirements
  - architecture.md — System-Design-Entscheidungen
  - api_spec.md — API-Contracts
  - images/ — Screenshots, Mockups, Wireframes
    - dashboard_mockup.png
    - flow_diagram.png
- src/ — Dein eigentlicher Code

Claude Code

Installation and Setup

Demo

Create PRD.md

- Product requirement document

Start Claude

- claude

Implement first version

- /init
- Implement @ai_docs\PRD.md

Claude Code

Claude.md

Eine Markdown-Datei im Projekt-Root mit wichtigem Context für Claude:

Typische Inhalte

- Build Commands
- Code Style Guidelines
- Architektur-Übersicht
- Testing-Strategien

Beispiel

```
# Build Commands
npm run build

# Code Style
Use TypeScript
```

Tipp: CLAUDE.md ins Git einchecken!

Claude Code

Slash-Commands

Vordefinierte Shortcuts für häufige Aufgaben

Session Management

- **/clear** - Context zurücksetzen
- **/compact** - Context komprimieren
- **/resume** - Session fortsetzen

Information

- **/help** - Alle Commands
- **/stats** - Nutzungsstatistik

Utilities

- **/agents** - Sub-Agents verwalten
- **/export** - Conversation exportieren

> **Tipp:** /help zeigt alle verfügbaren Commands!

Claude Code

Slash commands

/clear

Context zurücksetzen zwischen Tasks

/compact

Conversation-History komprimieren

/resume <n>

Frühere Session fortsetzen

/context

Context-Status und Verbrauch anzeigen

/agents

Spezialisierte Sub-Agents erstellen und verwalten

Claude Code

Subagents & Agent Teams

What are Subagents?

Isolated Claude instances that run tasks in parallel with their own context window.

- Delegate tasks (research, testing)
- Own context — no main session bloat
- Returns summary when done

Agent Teams

- Multi-agent collaboration
- Inherit permissions & MCP
- Parallel development workflows

Claude Code

Hooks

Scripts that fire automatically at lifecycle events — deterministic control, not AI.

Available Events

- PreToolUse / PostToolUse
- SessionStart / SessionEnd
- PromptSubmit / Compaction
- SubagentStart / SubagentStop

Use Cases

- Auto-lint after every file edit
- Run test suite after code changes
- Block dangerous commands (e.g., rm -rf)
- Send notifications when tasks complete

Claude Code

MCP Integration

Model Context Protocol (MCP) — an open protocol that connects Claude to external services and tools.

How it works

- Servers expose tools and resources
- Claude Code acts as MCP Client
- Configured via .mcp.json

Example Servers

- GitHub — PRs, issues
- PostgreSQL — DB queries
- Playwright — browser testing
- Google Workspace — Docs, Mail

Claude Code

Skills & Plugins

Skills

Reusable knowledge and workflows that Claude loads on demand — like macros for AI.

- SKILL.md files with metadata and instructions
- Auto-detected by context or invoked via /skill-name
- Reference skills (knowledge) or Action skills (workflows)

Plugins

- Bundle skills, hooks, subagents, and MCP servers into one installable unit
- Namespaced to avoid conflicts (e.g., /my-plugin:review)
- Installable via marketplace or local paths

Claude Code

Checkpoints & Sandboxing

Checkpoints

- Auto-saves code state before each change
- Rewind with Esc+Esc or /rewind command
- Restore code, conversation, or both
- Enables ambitious refactors with confidence

Sandboxing

- Filesystem isolation for project files
- Network isolation via proxy server
- OS-level enforcement (bubblewrap / seatbelt)
- Reduces permission prompts by 84%

Agentic Coding Levels

Claude Code

The 7 Levels of Agentic Coding Mastery

Level 1 Executor · CLI Prompts

Level 2 Resident · Context Engineering

Level 3 Specialist · Skills & Hooks

Level 4 Connector · MCP Servers

Level 5 Supervisor · Subagents

Level 6 Manager · Agent Teams

Level 7 Architect · CI/CD Integration

Claude Code

Levels 1–2: Foundation

Level 1: Executor · CLI Prompts

The Core Foundation

- Treat Claude Code like a powerful terminal
- One-off tasks: claude "fix the bug in auth.py"
- 200k+ token context window for instant project awareness
- No copy-pasting needed — reads local files dir ectly

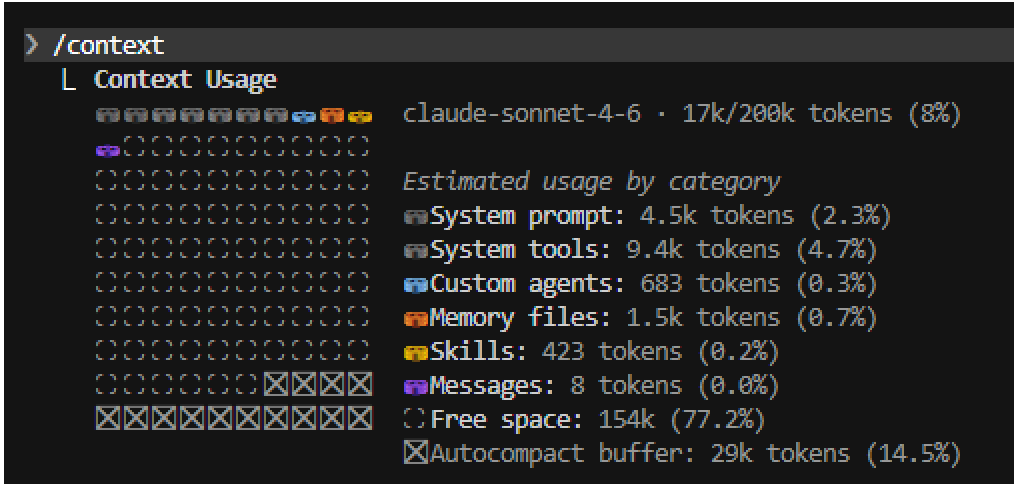
Claude Code

Level 2: Context Management

Level 2: Resident · CLAUDE.md

Personalized Memory and Context Management

- CLAUDE.md = System Prompt for your repo
- Define project rules, naming conventions, preferred libraries
- Read at every session start — no repeated instructions
- Transforms generic AI into a customized teammate
- Understand the importance of context



Output of /context

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CLAUDE.md und MEMORY.md im Vergleich.

CLAUDE.md

- Autor: Du
- Zweck: deine Anweisungen
- Geladen: zu Session Beginn

MEMORY.md

- Autor: Claude
- Zweck: eigene Notizen von Claude
- Geladen: zu Session Beginn

Agents

Context Engineering: From Prompts to Systems

Prompt Engineering

Fokus: die richtigen Worte finden

- Einzelne Prompts optimieren
- One-Shot Klassifikation / Textgenerierung
- Statisch: gleicher Prompt pro Aufruf
- Kein Memory, keine Tools

Context Engineering

Fokus: den gesamten Context optimieren

- System Prompt + Tools + Memory + History
- Dynamisch: Context pro Schritt kuratiert
- Tool-Ergebnisse, Files, Retrieval
- KV-Cache, Compaction, Compression

Sources: Anthropic (2025), Manus.im, Google ADK Documentation

Agents

Context Engineering: Practical Patterns

S
System Prompt
Instruktionen, Persona, Regeln, Output-Format

T
Tools
Funktionen die der Agent aufrufen kann (APIs, DB)

M
Memory
Chat-History
CLAUDE.md
MEMORY.md

R
Retrieval
RAG
Codebase-Index
Dokumentation

A
Agent Loop
Planning, Reflection
Tool-Aufrufe,
Observation

C
Compaction
KV-Cache
Kompression
Lost-in-the-Middle

Kernprinzip: Nicht den Prompt, sondern das gesamte Informationssystem um das Modell designen.

Claude Code

Levels 3–4: Automation & Integration

Level 3: Specialist · Skills & Hooks

Repeatability

- Skills: Markdown guides Claude invokes for co mplex workflows
- Hooks: Auto-trigger shell commands (e.g., npm test after edits)
- Stop explaining how — just tell Claude what to do
- Custom slash commands like /simplify or /debu g

Level 4: Connector · MCP Servers

External Integration

- MCP breaks Claude out of the local terminal
- Fetch data from Jira, Slack, Google Docs, data bases
- AI becomes environment-aware, not just code-aware
- Connect to your full ecosystem via MCP protoc ol

Claude Code

Levels 5–6: Multi-Agent Orchestration

Level 5: Supervisor · Subagents

Parallel Delegation

- Spawn multiple Claude subagents
- Each handles an independent task
- Parent coordinates & merges results
- Great for refactors & test suites

Level 6: Manager · Agent Teams

Structured Team Workflows

- Define roles: planner, coder, tester
- Agents hand off work in a pipeline
- Mimics real team dynamics with AI
- Scale beyond single-agent capacity

Level 7: Architect · CI/CD Integration

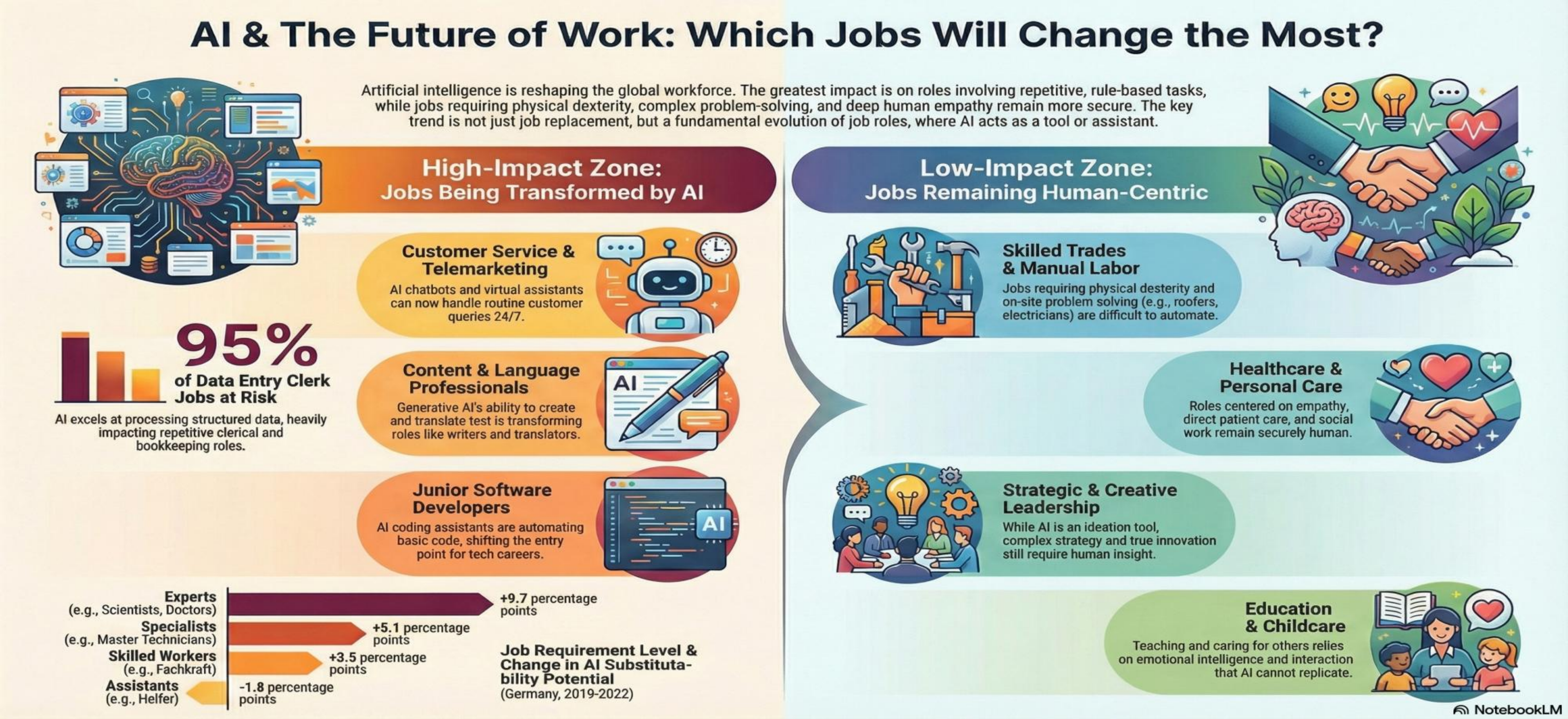
Autonomous Pipeline

- Claude runs headless inside CI/CD pipelines (GitHub Actions, GitLab CI)
- Triggered by events: new issues, PRs, scheduled jobs, deploys
- Zero human intervention — fully autonomous coding workflows
- Auto-generate PRs, run tests, fix failures, update documentation
- The ultimate goal: AI as a self-sustaining engineering partner

Socio-economic Impact, Risks & Chances

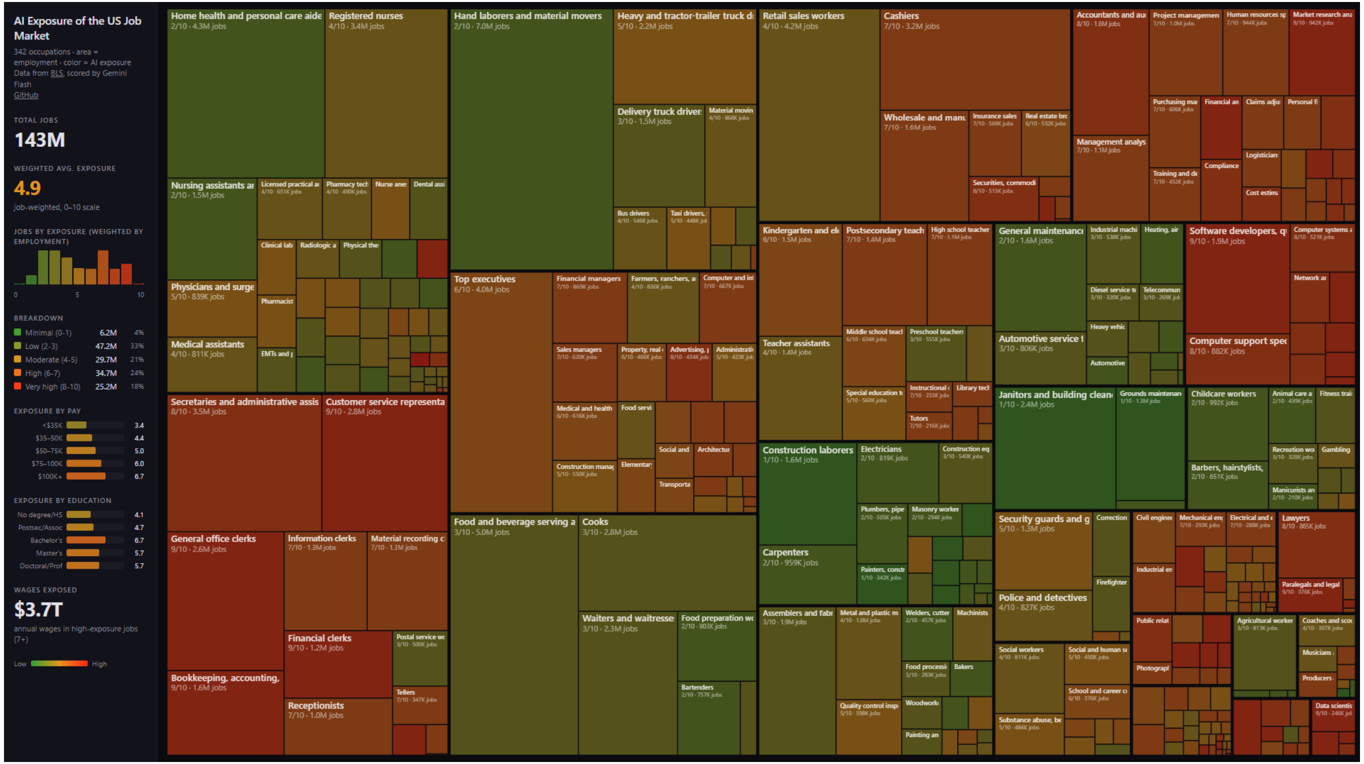
Socio-Economic Impact

Jobs



Socio-Economic Impact

Jobs – AI Exposure

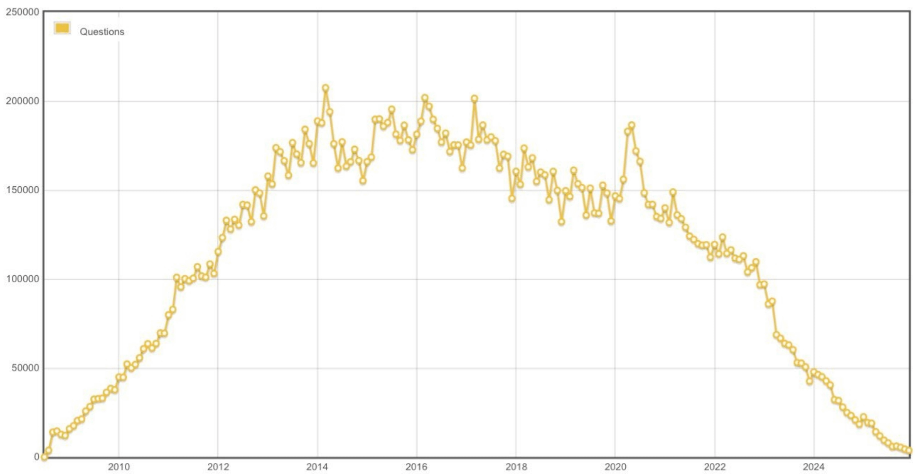


Source: Andrzej Karpathy, https://joshkale.github.io/jobs/

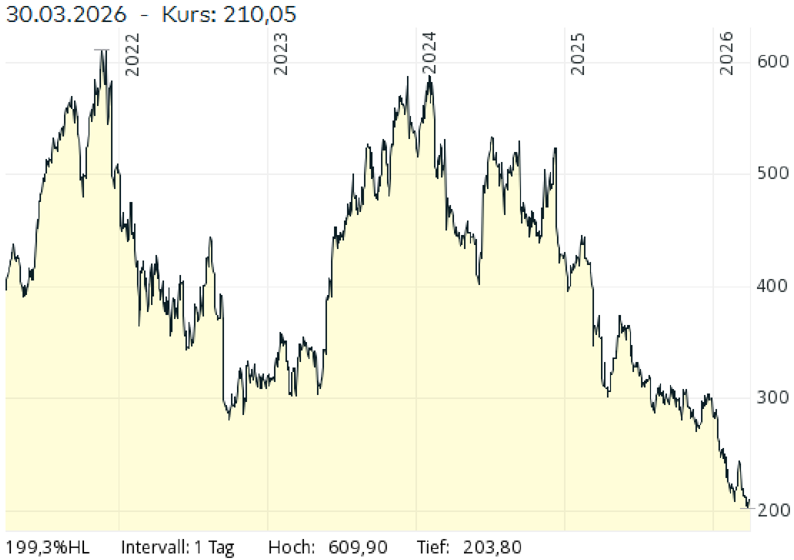
Socio-Economic Impact

Jobs – Impact on Software Development

Stackoverflow Use



Adobe Stockprice



Sources: https://www.reddit.com/r/singularity/comments/1q4llh1/stackoverflow_graph_of_questions_asked_per_month/

Comdirekt, Adobe stock price, 30.03.2026

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Agents

Security & Risks: Agentic Coding

Risks & Attack Vectors

Prompt Injection

- Schadhafte Anweisungen in Code-Kommentaren, READMEs, Dependencies

Supply Chain Risks

- Agent installiert unsichere oder halluzinierte Packages

Code Quality Erosion

- Code ohne Security-Review, Copy-Paste-Schwachstellen

Scaling Offense

- Gleiche Tools helfen auch Angreifern

Defense Improvements

Automated Code Review

- Jeder Entwickler kann Security-Reviews durchführen

Sandboxing & Isolation

- Filesystem- und Netzwerk-Isolation begrenzt den Agent-Radius

Continuous Hardening

- SAST, Dependency Scanning, IaC in CI/CD

Democratized Security

- Security-Wissen wird durch AI allen zugänglich

Source: Anthropic 2026 Agentic Coding Trends Report, OWASP LLM Top 10

Agents

Security: Guardrails & Best Practices

Permission System

Explizite Genehmigung für Writes, Shell, Netzwerk.
Granulare Policies pro Projekt.

Sandboxed Execution

Agents laufen in isolierten Containern/VMs.
Filesystem- und Netzwerk-Isolation.
OS-Level Enforcement.

Code Review Pipeline

Automatisierte SAST + Dependency Scan.
Agent-generierte PRs durchlaufen gleiche Reviews wie menschlicher Code.

Audit & Observability

Vollständiges Logging aller Agent-Aktionen.
Tracing für Prompts, Tool-Calls, Ergebnisse.
Kosten- und Latenz-Metriken.

Sources: OWASP LLM Top 10, Anthropic Sandboxing Docs, Apiiro Security Research

Summary

Was Sie mitnehmen sollten

Fünf Botschaften und ein Auftrag

1. Agentic Coding ist nicht Autocomplete 2.0

Agents planen, nutzen Tools, validieren ihr Ergebnis und iterieren — eine andere Klasse als Code-Vorschläge.

2. Disziplin schlägt Magie

Der Unterschied zwischen Vibe und Agentic Coding ist Workflow: Spezifikation, Validation Loop, surgical changes.

3. Context Engineering ist die neue Kernkompetenz

CLAUDE.md, Skills, Hooks, MCP — wer den Kontext des Agents gestaltet, gestaltet das Ergebnis.

4. Wählen Sie Ihr Tool nach Use Case, nicht nach Hype

CLI für Senior-Workflows, IDE für Refactoring, Browser für Prototyping. Es gibt nicht das eine richtige Werkzeug.

5. Die Levels sind eine Reise, kein Sprung

Starten Sie bei Level 1 (Executor) und arbeiten Sie sich hoch — jede Stufe baut auf der vorherigen auf.

Ihr Auftrag für nächste Woche

Installieren Sie Claude Code (oder ein Äquivalent), schreiben Sie eine CLAUDE.md für ein echtes Projekt und lassen Sie den Agent eine kleine Aufgabe erledigen. Lernen passiert beim Tun.

Claude Code

Best Practices

Context Management

Nutze **/clear** häufig zwischen Tasks

Mehrstufiger Workflow

Research → Plan → Implement → Test → PR

CLAUDE.md pflegen

Dokumentiere Commands, Style, Testing

Custom Commands sparsam

Nur für häufige Workflows - Natural Language bevorzugen

Sub-Agents für große Tasks

Spezialisierte Agents für komplexe Multi-Step-Ops

Claude Code

Summary

Claude Code ist...

- Agentisches Coding-Tool
- Direkt im Terminal
- Mit Projekt-Context
- Erweiterbar durch MCP
- Anpassbar durch Commands

Key Features

- Natural Language Interface
- Slash Commands
- CLAUDE.md für Context
- Sub-Agents
- Skills für Automatisierung

Starte mit **claude** im Terminal!

Dokumentation: code.claude.com/docs

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AI-Training

AI-Consulting

AI-Development